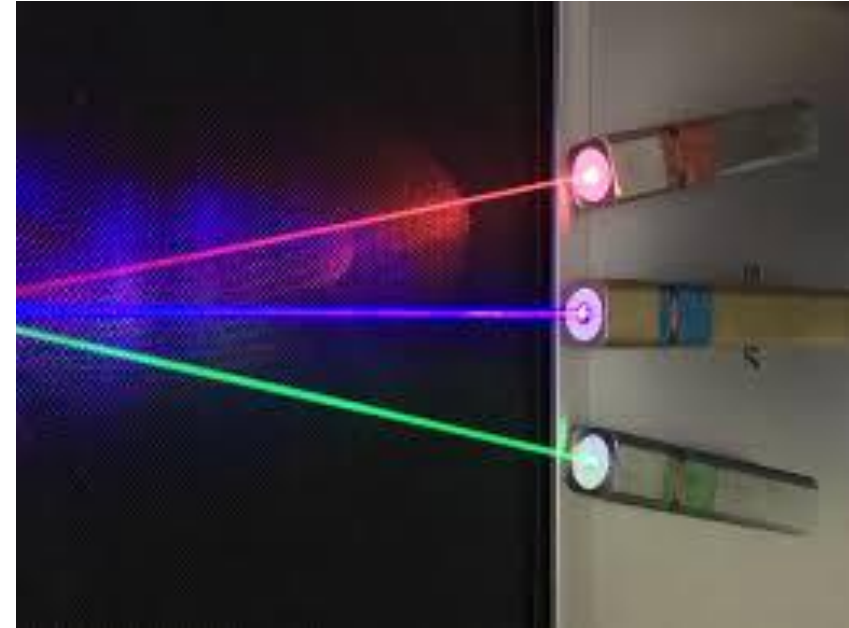


Module-1: LASER

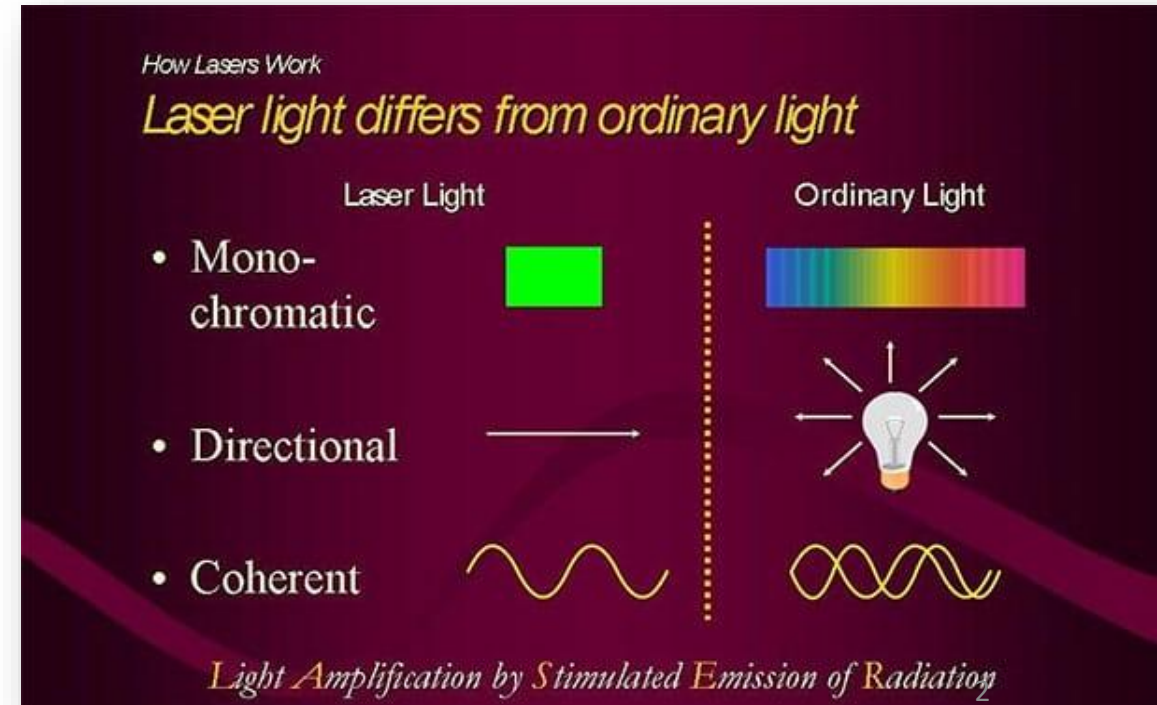
**How is LASER
different from regular
light?**



**Lasers and Optical Fibers
(Backbone of Modern Communication System)**

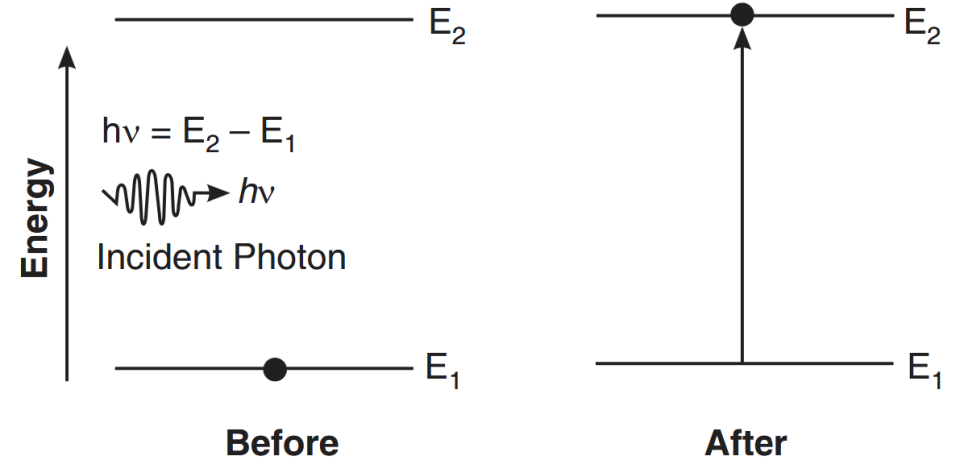
Introduction

- A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation
- The term “**LASER**” originated as an acronym for “*Light Amplification by Stimulated Emission of Radiation*”
- **LASER**
 - Produces a narrow, intense beam of light
 - Monochromatic
 - Coherent
 - Highly directional
 - Polarized (except for few laser's)



Quantum Processes

1. Absorption
2. Spontaneous Emission
3. Stimulated Emission

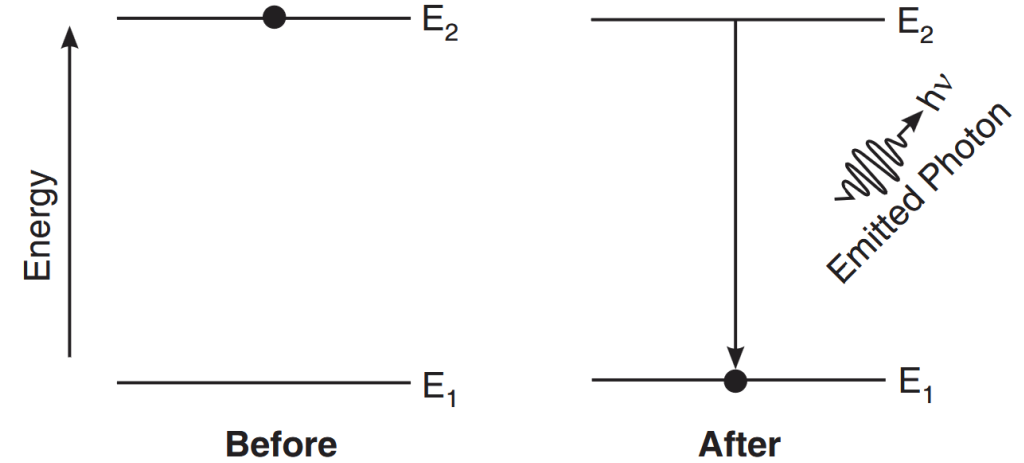


Process of Absorption of photons

Absorption:

- By absorbing a photon, an atom moves to a higher energy level through absorption

Quantum Processes

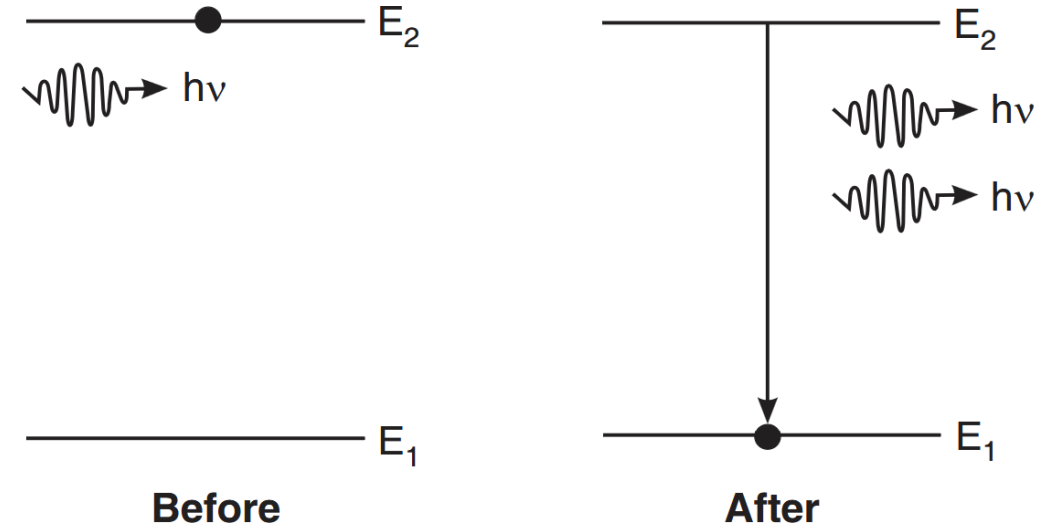


Process of Spontaneous emission of photons

Spontaneous Emission:

- Emission of photon by an atom without any external force is called Spontaneous emission
- The probability of spontaneous emission increases rapidly with energy difference between two energy states

Quantum Processes

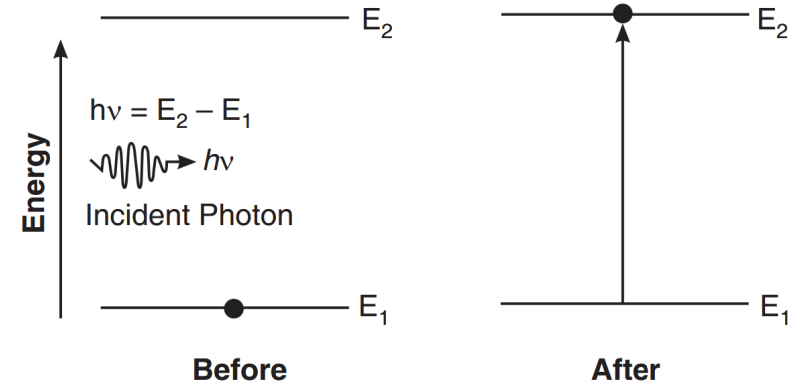


Process of Stimulated Emission of photons

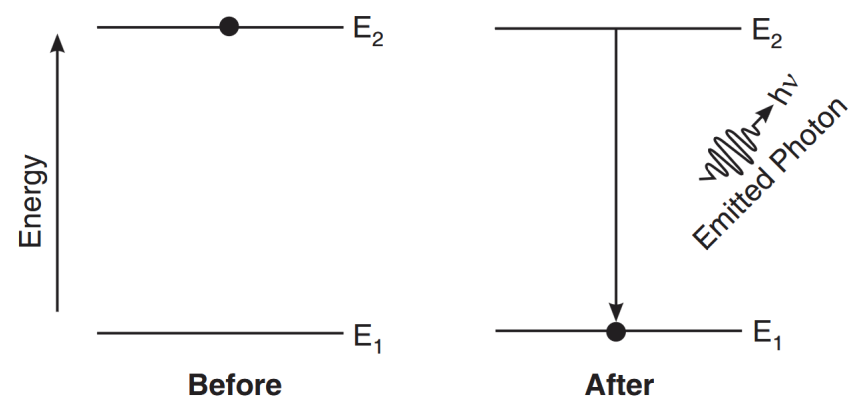
Stimulated Emission:

- The phenomenon of forced photon emission by an excited atom due to action of external agency (photon) known as stimulated emission
- The incident photon gets coherently amplified in this process and it is called amplification of light by stimulated emission of radiation

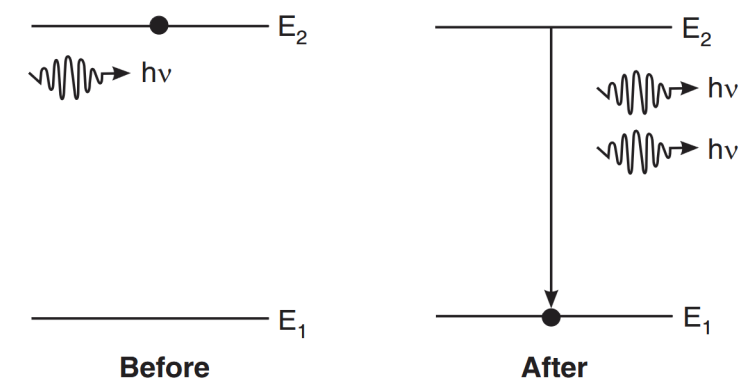
Recall



Process of Absorption of photons



Process of Spontaneous emission of photons



Process of Stimulated Emission of photons

Absorption: By absorbing a photon, an atom moves to a higher energy level through absorption

Spontaneous Emission: Emission of photon by an atom without any external force is called Spontaneous emission

Stimulated Emission: The phenomenon of forced photon emission by an excited atom due to action of external agency known as stimulated emission

Population Inversion

- Population inversion is a process of achieving more no. of **atoms, ions, molecules or free electrons** in the higher energy state as compared to the lower energy state
- To achieve population inversion, we need to supply energy to the **active medium**
- **Pumping:** The process of raising atoms from lower energy state to the higher energy state by an external source is called pumping
 - e.g. Optical, Electrical and Chemical pumping

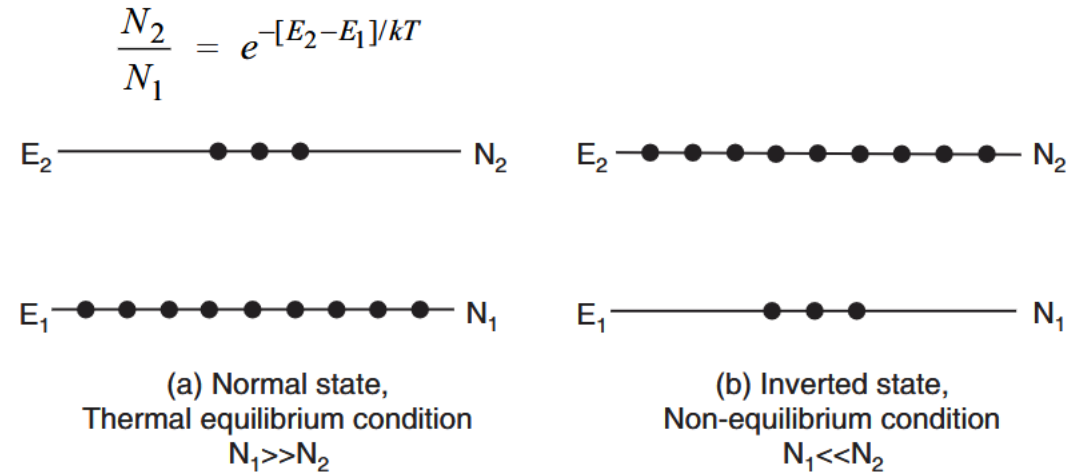
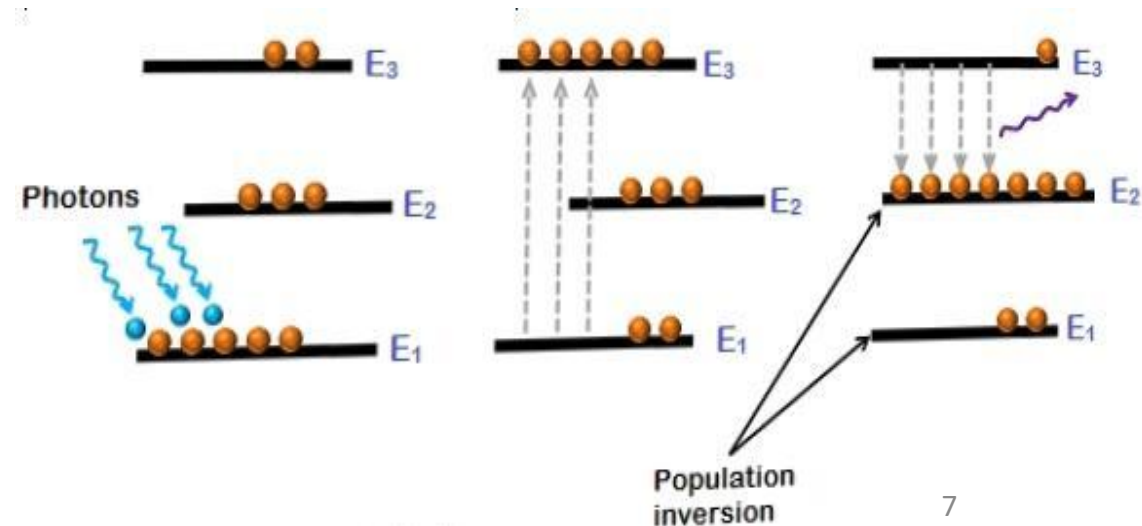
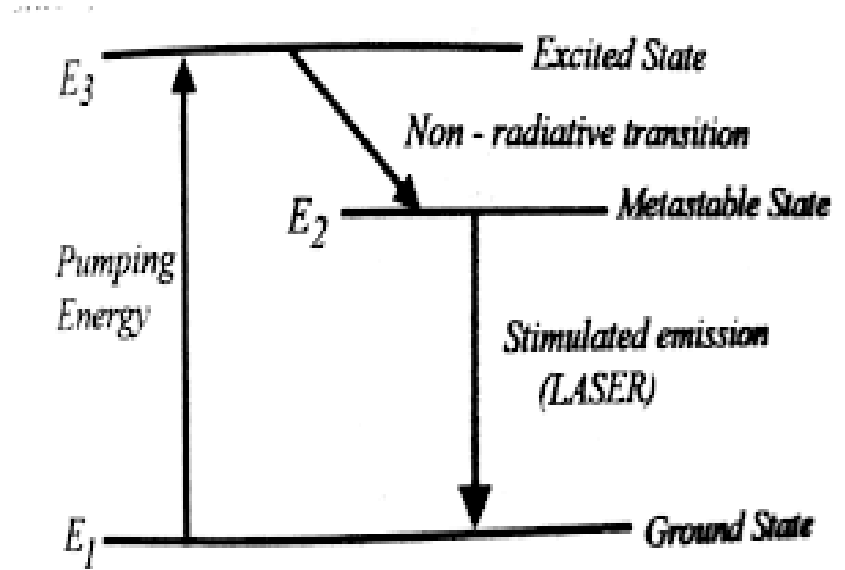


Illustration of Population Inversion (a) Normal state (b) Inverted state



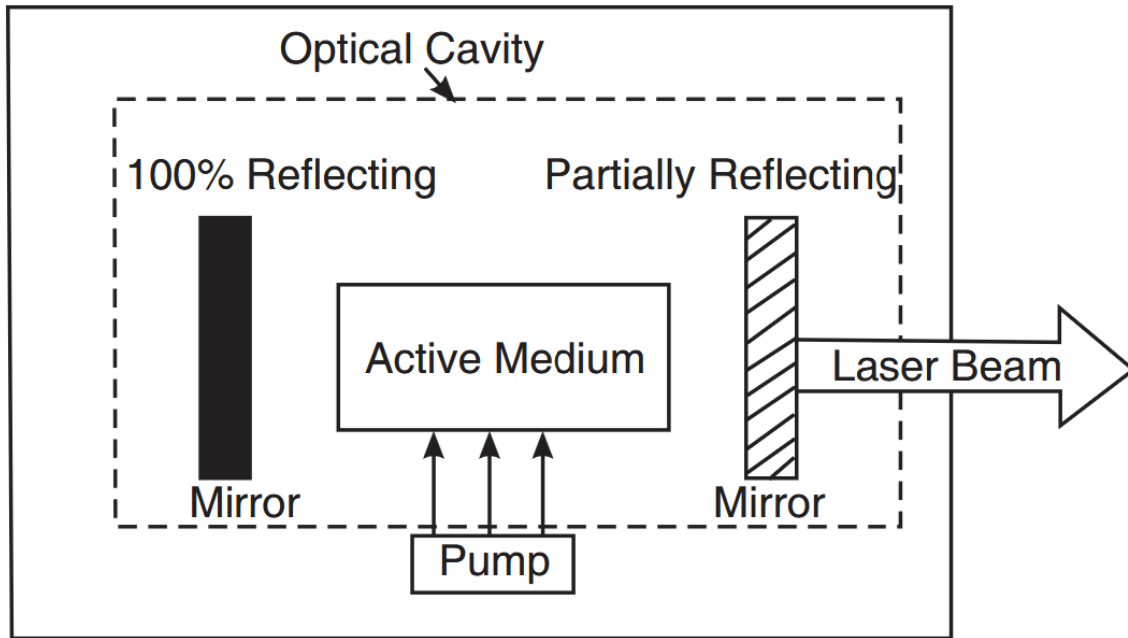
- **Optical pumping:** Optical pumping is a process in which light is used to raise atom, from a lower energy state to a higher energy state
- **Meta Stable State:** Meta stable state is a temporary excited state where atoms remain slightly longer as compared to the excited state



Active Medium and Active Centers

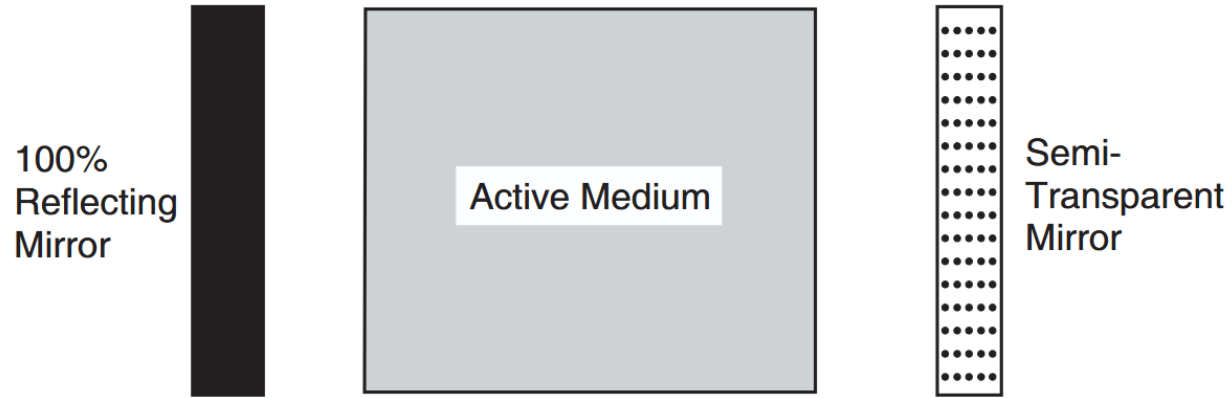
- A medium in which the light gets amplified is called active medium
 - Active centers are typically atoms, ions, molecule and free electrons of the active medium
 - Solid: Ruby Laser, Nd-YAG Laser
 - Liquid: Dye Laser
 - Gas: He – Ne Laser

Laser Components & Lasing Action

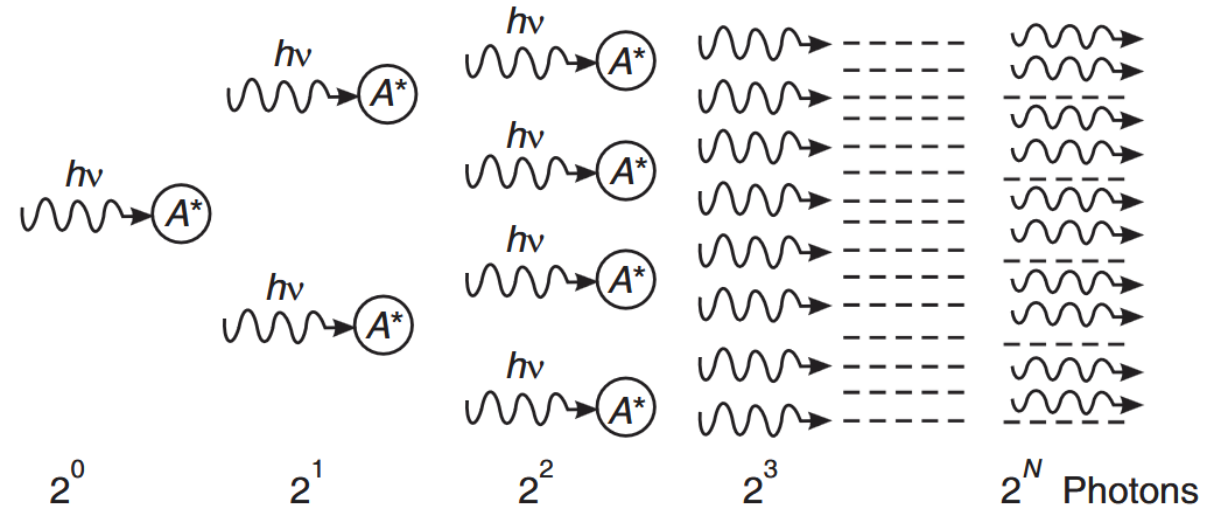


Components of a Laser

Resonant Cavity



Coherent amplification of light



Multiplication of Stimulated photons into an avalanche

Coherence Length & Coherence Time

Coherence length:

- The distance over which the laser light remains coherent i.e. the light waves stays in phase
 - Generally used in Holography and Interferometry

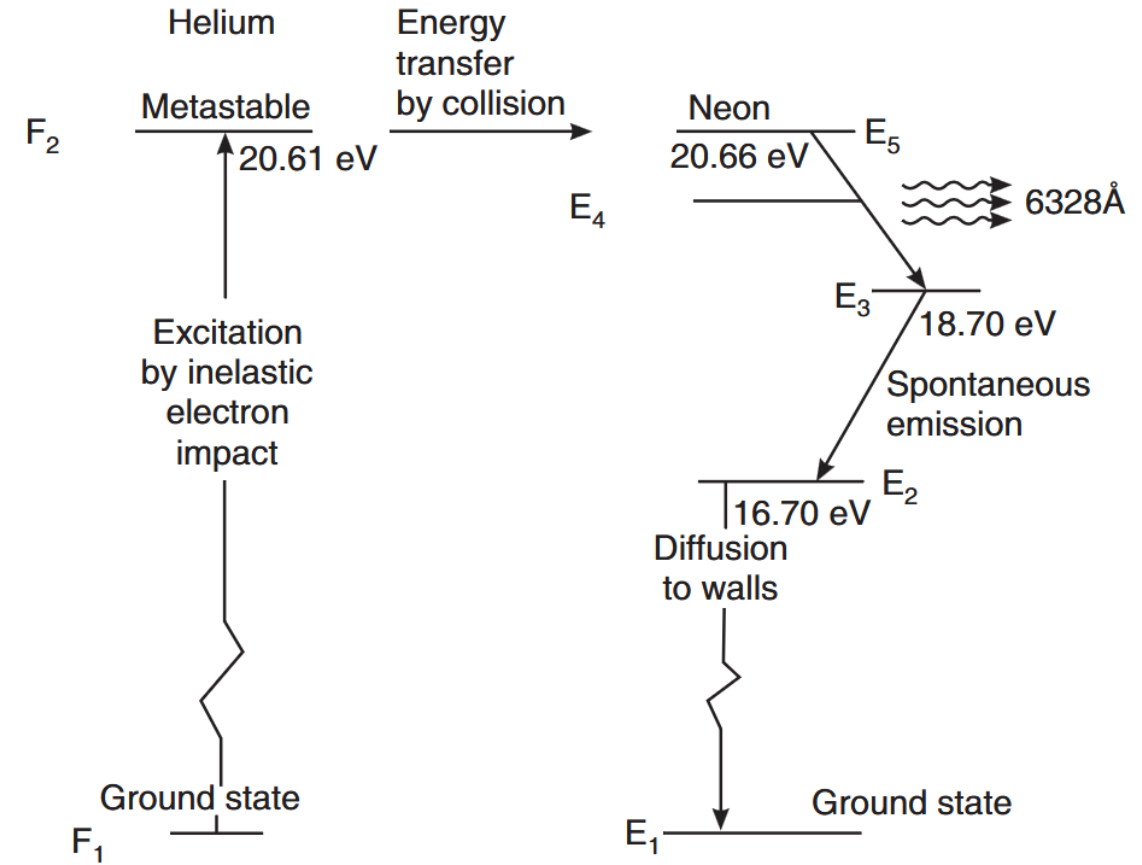
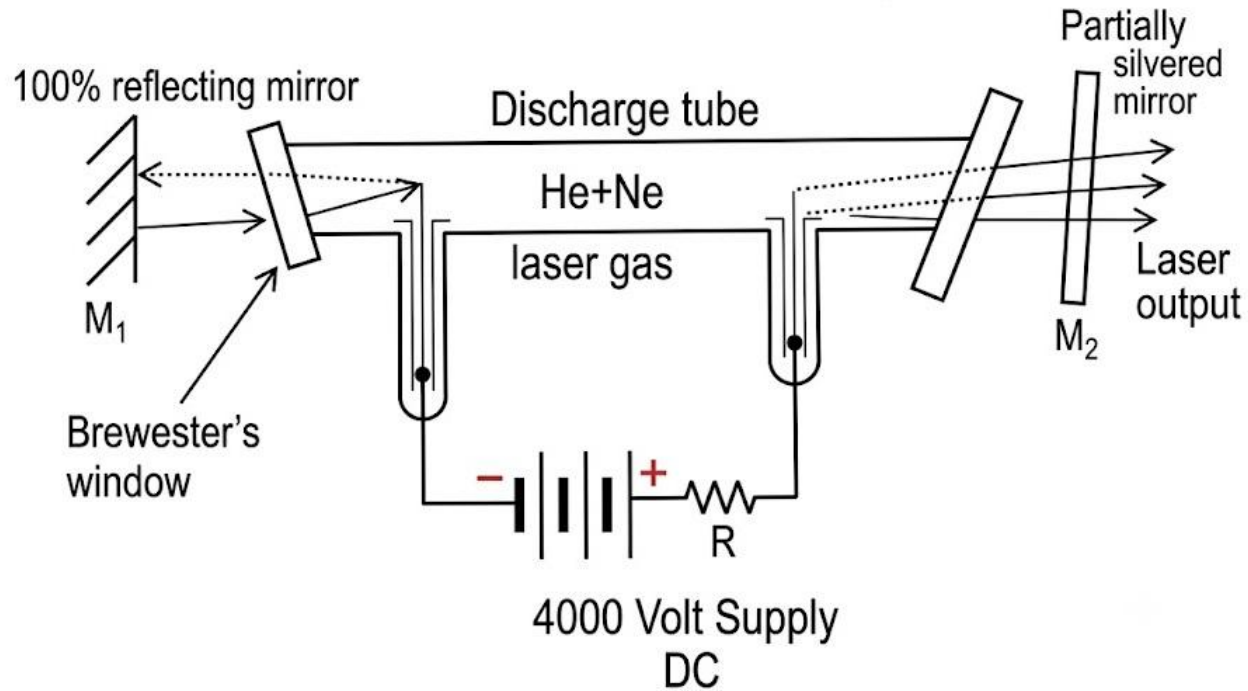
Coherence time:

- The time over which the phase of the laser light waves remains constant
- *The coherence length and coherence time measures the laser's phase stability and beam quality*

Characteristics of Laser's

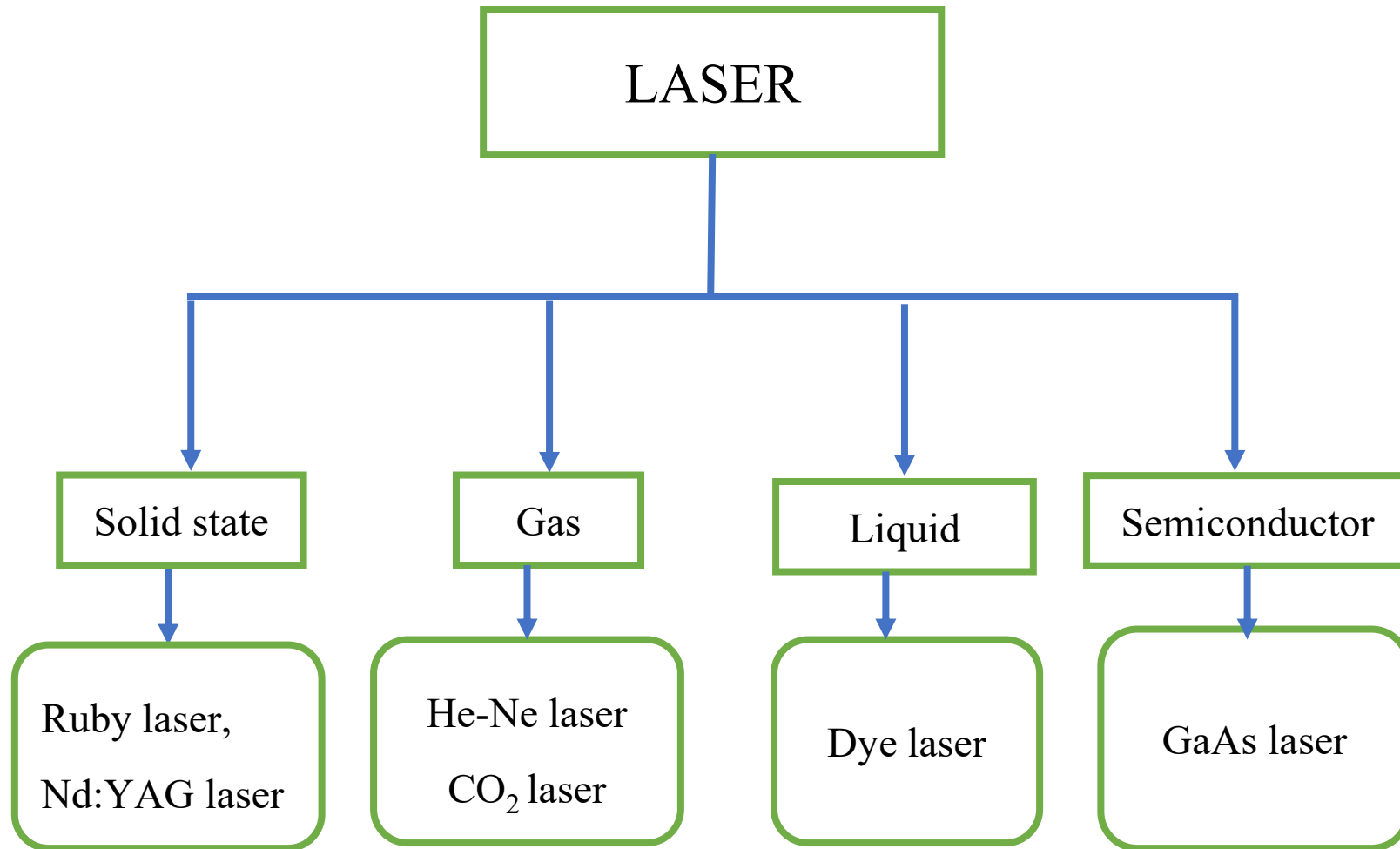
1. **Coherence:** *All the light waves travel in the exact same phase, both in time (temporal coherence) and space (spatial coherence). This means the crests and troughs of the light waves line up perfectly.*
2. **Monochromaticity:** *Laser light has a single, pure color or wavelength because the photons are emitted with identical energy and frequency*
3. **Directionality:** *A laser beam travels in a single, straight path with very little spreading or divergence. It can cover long distances while staying focused in a tiny spot*
4. **High Intensity:** *Because the light energy is packed into a narrow beam and a small area, a laser has extreme brightness and power density*

He – Ne LASER (Construction & Working)



Energy levels of helium and neon and Laser transitions.

Types Of LASER



Note:

- Nd: Neodymium , YAG: Yttrium aluminum Granet (Crystal)
- GaAs: Gallium Arsenide

1. Elementary Knowledge of LiDAR

LiDAR: Light Detection And Ranging

➤ *LiDAR is a remote sensing technology that uses laser light to measure the distance between a sensor and an object*

➤ **How does LiDAR work ?**

1. Emit → Laser pulse is sent towards an object
2. Reflect → Light reflects back from the object
3. Detect → Sensor detects the reflected pulse
4. Calculate → Distance is calculated as $d = (c \cdot \Delta t) \div 2$

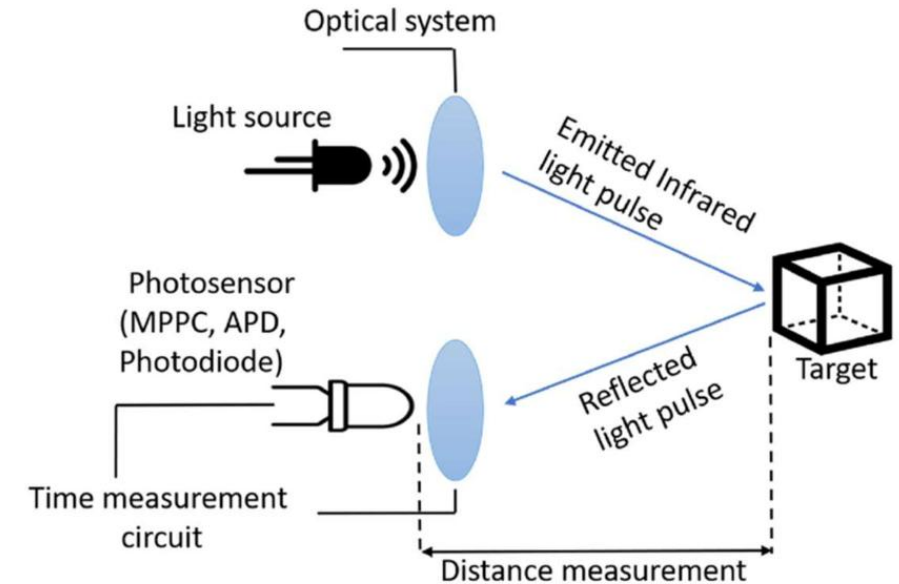
d = distance to the object

c = speed of light

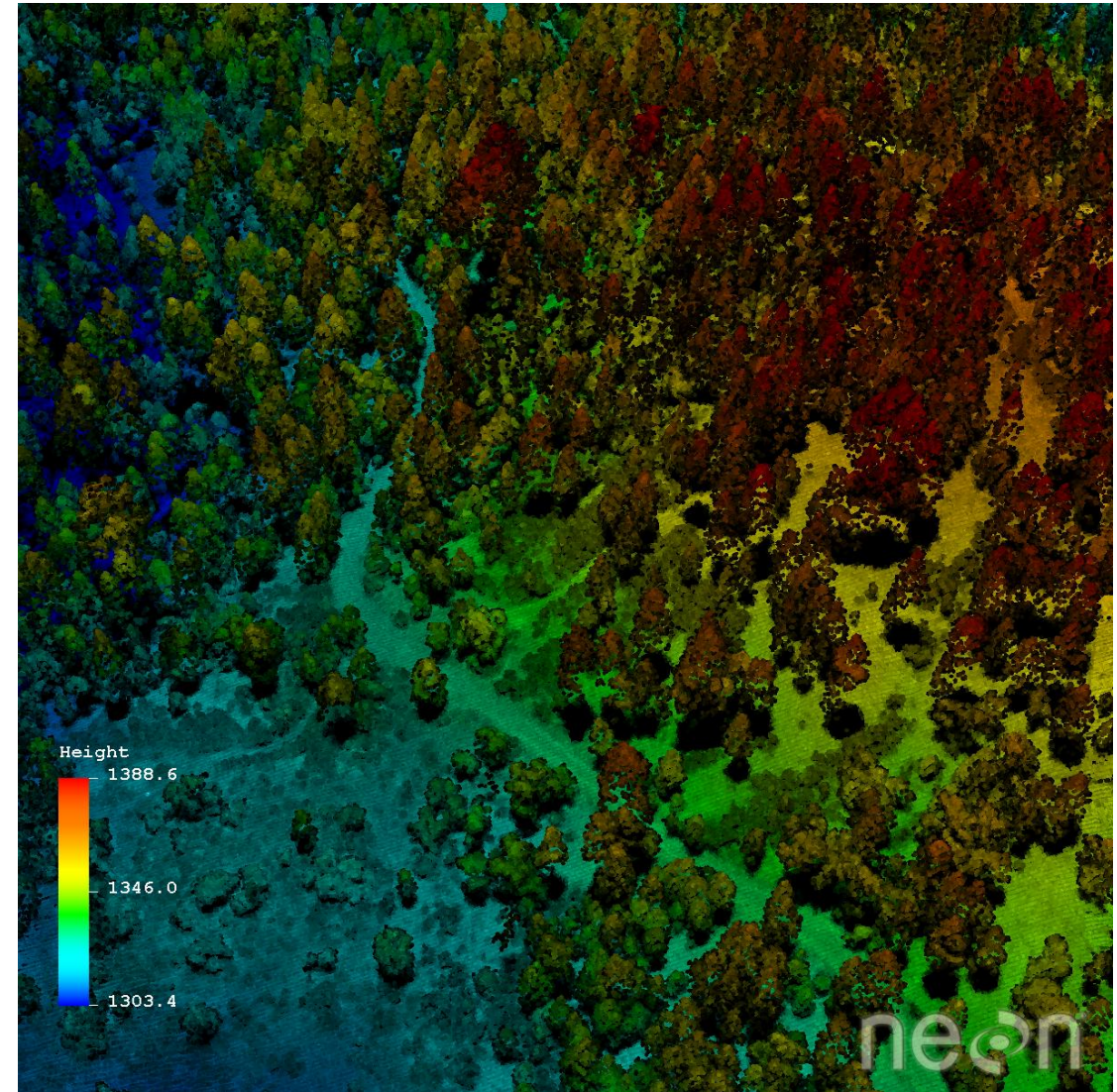
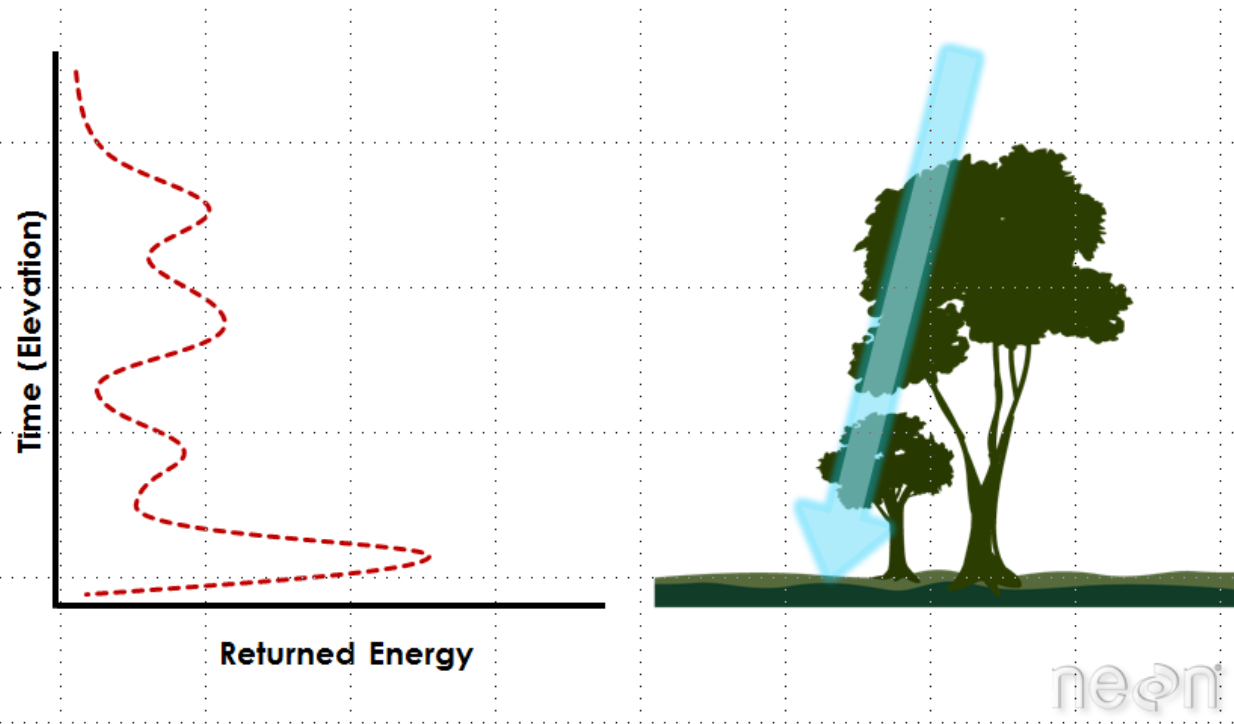
Δt = round-trip time of the laser pulse

➤ **Applications:**

🚗 Autonomous vehicles | 🗺️ 3D Mapping | 🌲 Forestry



A Complex LiDAR Application



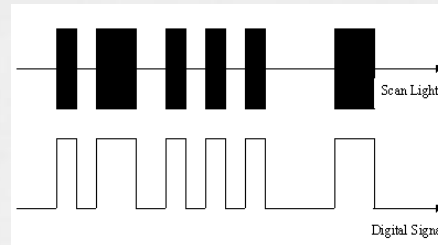
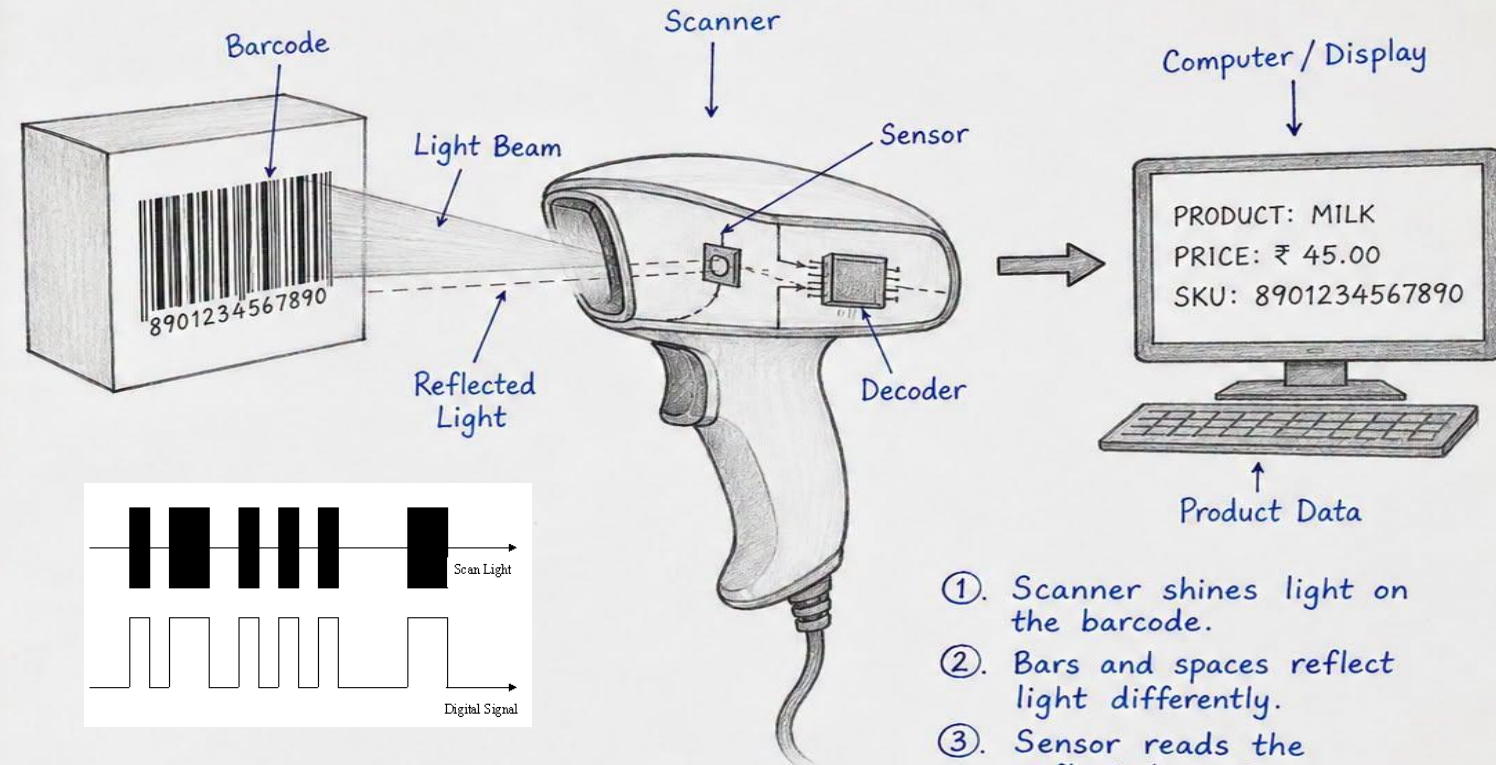
Author: *Leah A. Wasser*

2. Barcode Reader

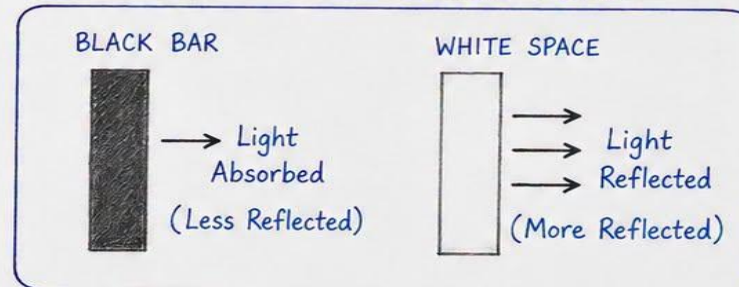
Applications:

1. *Retail billing*
2. *Inventory management*
3. *Libraries*
4. *Logistics*
5. *Product identification*

HOW A BARCODE SCANNER WORKS



HOW BARS AND SPACES WORK



- ①. Scanner shines light on the barcode.
- ②. Bars and spaces reflect light differently.
- ③. Sensor reads the reflected pattern.
- ④. Decoder converts pattern into numbers.
- ⑤. Computer finds the product information.
- ⑥. Data appears on the screen.



A barcode stores information using patterns of bars and spaces.

3. Metal Works



Industrial & Manufacturing Applications:

- LASER Cutting
- LASER Welding
- LASER Engraving & Marking

General Uses of LASER

- **Medical and Healthcare:** Ophthalmology (eye), Dermatology (skin), Surgery etc.
- **Communication & IT:** Fiber Optic Networks, Data transport
- **Metrology, Measurement & Défense:** Distance measurement, Military equipment's
- **Entertainment:** Displays, LASER show using drones
- **Research:** Identification of chemical composition, study microorganism

